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Paper:

Real-Time Decision Making for Autonomous City Vehicles

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This paper addresses the topic of real-time decision making by autonomous city vehicles. Beginning with an overview of the state of research, the paper presents the vehicle decision making & control system architecture, explains the subcomponents which are relevant for decision making (World Model and Driving Maneuver subsystem), and presents the decision making process. Experimental test results confirm the suitability of the developed approach to deal with the complex real-world urban traffic.

Keywords: autonomous city vehicles, real-time decision making, intelligent control

1. Introduction

Autonomous city vehicles are expected to significantly improve the safety, efficiency, energy consumption, pollution, comfort, and mobility [1–3].

Ideally, such vehicles will be able to operate without requiring major road infrastructure changes, and will be able to cope with the urban traffic complexity, sharing the roads with human-driven vehicles, and pedestrians [4].

The efficiency with respect to travel time and energy consumption will be achieved using Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication and cooperation, which will enable autonomous city vehicles to automatically avoid traffic congestions, to cross intersections faster and safer, to avoid unnecessary stops, and to drive autonomously to and from parking areas. By replacing the human driver with an intelligent control system, the high number of road accidents caused by human errors, driver fatigue, distractions, or speeding, will be eliminated.

Another future vision are fleets of autonomous city vehicles, so called Cybernetic Transport Systems (CTS), which will combine the benefits of public transport (e.g., accessibility for all citizens) with those of today's cars [2, 5].

The remainder of this paper is structured as follows. The rest of this section gives an overview of the state of research, Section 2 presents the control system architecture, Section 3 explains the World Model, Section 4 ex-

plains the driving maneuver concept, Section 5 presents the developed decision making approach, and Section 6 presents the experimental decision making tests. Section 7 concludes the paper.

Research Review

Despite much research related to autonomous city vehicles, the topic of real-time decision making, especially focusing on the high requirements of non-simplified real-world urban traffic conditions, is a relatively recent research topic.

In an early example in 2002, Australia's Griffith University Intelligent Control Systems Laboratory (ICSL) and France's INRIA-IMARA Laboratory showed how cooperative autonomous vehicles followed traffic lanes, avoided static obstacles, detected dynamic obstacles, and performed driving maneuvers which required cooperation, such as traversing unsignalized intersections or predicting the driving maneuvers of other autonomous vehicles [4, 6]. However, at that time, the performed driving maneuvers were activated manually, since a real-time decision making module was yet to be developed. The continuation of the project resulted in the development of the real-time decision making solution presented here.

The first experimental Cybercar (fully automated vehicle) was prototyped in 1997 and demonstrated at the airport in Amsterdam [7], while the world's first publicly accessible Cybercar was developed in Rotterdam (Rivium project) as part of the European CyberCars/CyberMove initiative. In order to avoid legal difficulties, the Cybercar vehicles have been so far operated in restricted environments, such as airports or theme parks [2, 7].

The CyberCars/CyberMove project (2004) was followed by Cybercars2 (started in 2006) and CyberC3 (2004), which focused on marketing European CyberCars in China [8]. In these projects, the vehicles' positions were determined by the guidance module using road magnets, and using vision and laser sensors. However, since the vehicles were not able to leave the magnetic path, they did not require real-time decision making.

The major recent event demonstrating the state of the art autonomous city vehicle technology was the DARPA Urban Challenge 2007 [9], an autonomous vehicle race in a simulated urban environment. Although DARPA tried to create an environment similar to urban traffic by including

human-driven vehicles, the traffic environment conditions were simplified compared to real urban traffic (e.g., no pedestrians or bicycles).

As a result, it was possible to successfully complete the race using LIDAR sensor information alone, and without recognizing and distinguishing among obstacle types. Some teams even decided a short time before the race not to use their already fully developed computer vision systems for obstacle detection (e.g., vehicles Odin [10] and Talos [11]). With very few driving maneuvers, such as following a road, crossing intersections, and driving on parking areas, it was possible to successfully complete the race [12].

These simplified road traffic conditions resulted in very limited requirements for real-time decision making. Although successful in the Urban Challenge, some teams admit that their approaches are not sufficient to cope with real-world urban traffic requirements. For example, Junior's development team states that due to the simplified conditions, their vehicle, and probably any other vehicle competing in this race, would probably not be able to cope with a realistic city traffic environment [13].

The DARPA Urban Challenge 2007 winners, the Tartan Racing team (vehicle Boss), acknowledged that their traffic representation was not sufficient to make intelligent driving decisions compared to human drivers, noting that *"A richer [traffic] representation including more semantic information will enable future autonomous vehicles to behave more intelligently"* [12].

2. The Autonomous Vehicle Decision Making & Control System Overview

The autonomous vehicle control system consists of the following functional subsystems (**Fig. 1**):

- Perception subsystem,
- Real-Time Decision Making & Driving Maneuver Control subsystem,
- Driving Maneuvers subsystem,
- Vehicle Interface subsystem.

The purpose of the Perception subsystem is to collect available information about the vehicle's road traffic environment, to manage and process it, and to provide it in an adequate form to the Real-Time Decision Making & Driving Maneuvers subsystem.

Based on the information provided by the Perception subsystem, the Real-Time Decision Making & Driving Maneuver Control subsystem makes driving decisions. This software subsystem decides about the activation and the execution of the most appropriate driving maneuver.

The Driving Maneuvers subsystem contains a set of closed-loop control algorithms, each able to maneuver the vehicle in a specific traffic situation. The Driving Maneuvers direct their output to the Vehicle Interface subsystem.

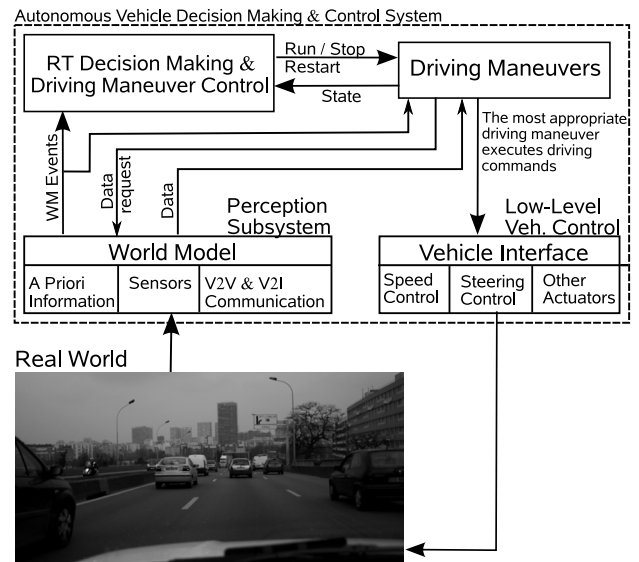


Fig. 1. Simplified view of the autonomous vehicle decision making & control software architecture and the flow of data.

The Vehicle Interface subsystem contains hardware and software components, which control the vehicle's speed, steering angle, and other actuators (e.g., transmission).

3. The World Model

For the purpose of decision making, various types of input information about the vehicle's traffic environment are required, such as a priori information, information obtained from on-board sensors in real-time during the vehicle's movement, and information obtained through communication (**Fig. 2**).

The a priori information, which is available preceding the autonomous vehicle's operation, includes for example the planned travel path, coordinates of intersections and roundabouts, and/or other relevant information about the road infrastructure, such as the number of traffic lanes. This a priori information is specified for the planned route similar to the DARPA Urban Challenge 2007 (i.e., Route Network Definition File, RNDF) [9, 14].

Additionally, the autonomous vehicle continuously obtains real-time information about its traffic environment either from its on-board sensors (e.g., cameras, LIDAR, RADAR), through Vehicle-to-Vehicle (V2V) communication, or through Vehicle-to-Infrastructure (V2I) communication (e.g., a traffic management centre).

The purpose of the World Model is to merge this information and to provide at any given time an accurate and up-to-date representation of the autonomous vehicle's traffic environment, which is then used as input information for the Real-Time Decision Making subsystem.

Consequently, the World Model's most relevant functional requirements are to store a priori information, the information provided by sensors, the information obtained through communication with other vehicles or a

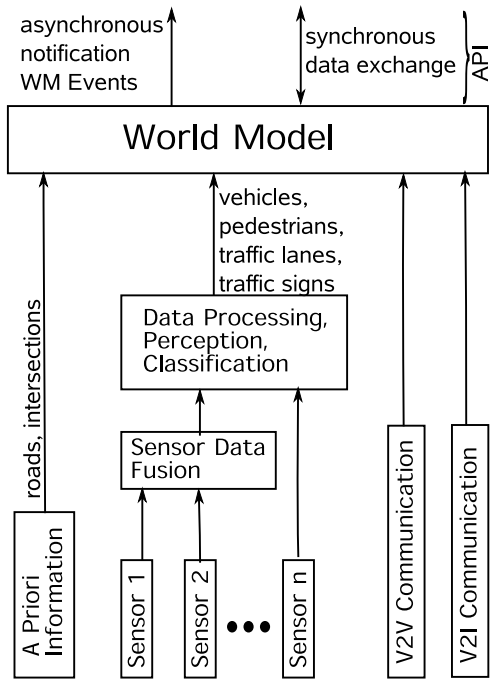


Fig. 2. World Model input and output.

traffic management centre, to merge and update the a priori information with the information obtained continuously from sensor and communication components, and to calculate the relationships among the stored entities. For example, if sufficient data is available, the World Model determines the positions of the current road, the current traffic lane, obstacles on the current road, the type of obstacles on the current lane/road, and distances to obstacles.

Additionally, the World Model notifies the Decision Making subsystem about relevant events in the traffic environment through an asynchronous mechanism, and provides complete access to all stored information in response to synchronous data requests, as detailed in [15].

4. The Driving Maneuver Subsystem

The highly complex problem of autonomous driving in the city environment is divided into several subtasks, the driving maneuvers, which are able to maneuver the vehicle in a specific traffic situation. Although this approach has been widely accepted and adopted by many research groups under different names (e.g., actions, behaviors, driving maneuvers, etc.) [16–18], detailed descriptions of the so far developed approaches have not been published.

We define driving maneuvers as closed-loop control algorithms, each capable of maneuvering the autonomous vehicle in a specific traffic situation. In order to enable the Decision Making subsystem to switch between execution of various maneuvers, all driving maneuvers are based on a common general structure. Their operational behavior is designed using *deterministic finite automata*

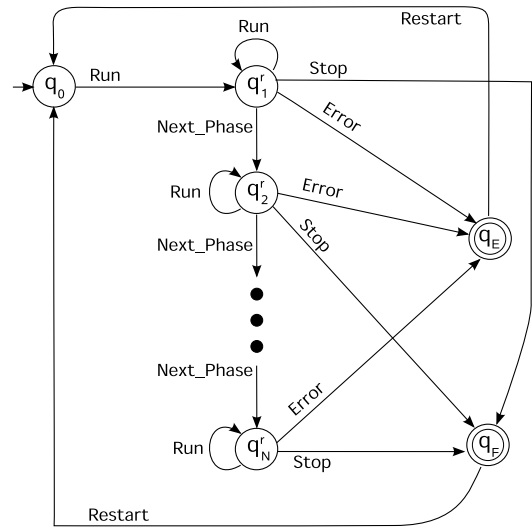


Fig. 3. A driving maneuver finite automaton with multiple Run states. The number of Run states $q_i^r, i = 1, 2, \dots, N$ equals the number of driving maneuver phases.

(deterministic finite state machines) as follows (Fig. 3):

- a *start state* q_0 ,
- 2 *final states* $\{q_F, q_E\} = F$,
- a set of *Run states* $Q^{run} = \{q_1^r, q_2^r, \dots, q_n^r\} \subset Q$,
- a set of input symbols Σ , consisting of at least the symbols:
Run, Stop, Restart, Error,
- the state transition function $\delta : Q \times \Sigma \rightarrow Q$, which defines the automaton's transitions between its states.

The start state q_0 is the *waiting* or *idle* state, in which the automaton is waiting for the *Run* signal from the Real-Time Control & Decision Making module.

The Run states $q_1^r, q_2^r, \dots, q_n^r$ perform the maneuvering of the vehicle. Each of them includes checking of necessary preconditions, such as the availability of World Model information and safety conditions. As long as the preconditions are met, the Run states execute control algorithms. Otherwise, if certain preconditions are not met, the vehicle is stopped, the *Error* symbol is generated, and the automaton enters the error state q_E .

A driving maneuver finishes in one of the final states q_F (finished) or q_E (error). The final state q_F represents a successful completion of the driving maneuver, while the error state q_E signals that the driving maneuver was aborted due to an error or some other reason. The information regarding the final state of a driving maneuver is used by the Real-Time Control & Decision Making module.

The set of input symbols mean the following:

$$\Sigma = \{Run, Stop, Restart, Error\}$$

- *Run*: request to begin the execution of the driving maneuver,

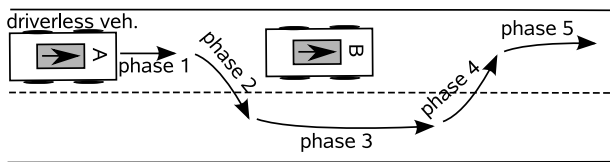


Fig. 4. The overtaking maneuver is decomposed into five phases. In the finite automaton each phase is represented by a Run state.

- *Stop*: request to stop the execution of the driving maneuver,
- *Restart*: request to restart the driving maneuver (i.e., to reset the automaton to its start state q_0),
- *Error*: some error occurred, which makes the continued execution of the driving maneuver impossible. The error can be for example due to one or more missing parameters necessary for the execution of the driving maneuver, or due to unfulfilled preconditions which are required for the successful execution of the driving maneuver.

This finite automaton based approach enables the systematic division of the relatively complex task of autonomous driving into subtasks with manageable complexity. Complex driving maneuvers are divided into maneuver phases (**Fig. 4**), where each phase is represented by a Run state which implements the closed-loop control algorithm.

5. The Decision Making Subsystem

An obvious and common approach to manage the complex control task of maneuvering an autonomous vehicle in the urban traffic environment is to divide the overall task into multiple independent subtasks, the driving maneuvers. Decision making in this context refers to the problem of identifying the most appropriate driving maneuver to be performed under the given road traffic circumstances.

The currently available solutions for this problem cannot deal adequately with the high complexity of real-world, non-simplified urban traffic conditions, where the complex decision making logic depends on a large variety of information about the traffic environment.

5.1. System Specification

Without the loss of generality, it can be assumed that the specification requires the decision making module to respond to certain traffic conditions in a defined way, i.e., certain traffic conditions imply the execution of certain driving maneuvers.

The occurrence of any traffic condition is represented by a set of discrete events, which can occur simultaneously, and which are detected by the vehicle's on-board

sensors or through V2V/V2I communication. These discrete events are sent to the decision making module by the World Model in the form of World Model events.

Based on the set of discrete World Model events, the system specification defines the required road traffic conditions in which a driving maneuvers can be safely performed (**Table 1**).

Consequently, the operation behavior of the decision making module is modeled in line with discrete event-driven systems [19]. Given the complexity of urban traffic, a complete specification for a typical decision making module will consist of a large number of defined inputs (i.e., discrete events), and a large number of outputs (i.e., driving maneuvers), which are logically interrelated in a complex way.

The decision making problem is specified as follows:

- a set $M_{all} = \{m_1, m_2, \dots, m_n\}$, $n \in \mathbb{N}$, of all available driving maneuvers which can be performed by the autonomous vehicle,
- a k -tuple $(w_1, w_2, \dots, w_k) \in W_{events}$ of World Model events, $w_l \in \{0, 1\}$, $l = 1, 2, \dots, k$,
- a route planner direction indication $d_i \in D_{route} = \{\text{forward straight, forward right, forward left, turn around}\}$.

The general problem of decision making in this context is to identify the most appropriate driving maneuver $m_{most_appr.} \in M_{all}$, which leads to an autonomous vehicle driving behavior conforming to the specification.

5.2. Decomposition into Subproblems

The decision making problem is decomposed into the following two consecutive stages:

1. Decision on the *feasible, safety-critical driving maneuvers* subject to World Model events and route planner indication. A driving maneuver is defined as feasible if it can be safely performed in a specific traffic situation, and is conforming to the road traffic rules. In any traffic situation, there can be multiple feasible driving maneuvers (e.g., overtaking a stopped vehicle or waiting for it to continue driving).
2. Decision on the most appropriate driving maneuver. This stage selects and starts the execution of one single driving maneuver, which is the most appropriate for the specific traffic situation. Only those driving maneuvers which have been selected as feasible (and therefore safe), are considered in this stage. Therefore, this stage is not safety-critical because it does not include any decision making attributes which affect safety.

The decomposition into two decision making stages leads to smaller subproblems of manageable complexity, enabling the verification and testing of each stage in particular. While the main focus of the first stage is to determine which driving maneuvers are safe and conform to traffic rules, the second, non safety-critical decision stage focuses on improving comfort and efficiency.

Table 1. Examples of discrete events of relevance to the selected driving maneuvers (*Overtake Right*, *Follow Lane* and *Stop&Go*).

Discrete Event	Explanation
w_1 : right lane boundary detected	The right lane boundary has been detected and its position is known
w_2 : left lane boundary detected	The left lane boundary has been detected and its position is known
w_3 : right lane boundary crossable	Changing lanes onto the right lane is possible
w_4 : right ongoing lane detected	The right lane has been detected
w_5 : obstacle on right ongoing lane	A moving or static obstacle has been detected on the right lane
w_6 : “no overtaking” sign detected	Overtaking is prohibited
w_7 : approaching intersection	The vehicle is approaching a close intersection
w_8 : any kind of obstacle in front	A static or moving obstacle has been detected in front
w_9 : moving vehicle in front	Another vehicle has been detected in front
w_{10} : static obstacle in front	A detected obstacle is blocking the lane
:	:

5.2.1. The First Decision Making Stage

The goal of the first decision making stage is to select the feasible driving maneuvers, i.e., the subset of all driving maneuvers, which can be performed without putting any traffic participants at risk.

The following aspects are relevant for the selection of feasible driving maneuvers:

- Information about the vehicle’s environment, which is provided in the form of World Model events.
- Knowledge about traffic rules and their compliance.
- Information about the planned travel direction, which is assumed to be provided by the route planner.

Each driving maneuver requires certain World Model information, which is provided by the World Model in the form of events. Therefore, occurring World Model events define which driving maneuvers are operational (i.e., which can be performed). In order to comply to traffic rules, additional restrictions of driving maneuvers are required. These are embedded in the first step of this stage (DMU1A).

As a third aspect, the planned traveling route plays a further role in reducing the number of candidate driving maneuvers. Driving maneuvers which lead the vehicle into a wrong direction, or maneuver it inadequately with respect to the planned route, are omitted from the set of feasible driving maneuvers. For instance overtaking a slower vehicle while the route planner indicates that a U-turn is needed, may not be adequate.

Consequently, the large number of factors to be considered in the first decision making stage requires a model which enables the design and analysis of a highly complex operational behavior. As Petri nets [20] are a suitable modeling method for this purpose, a Petri net is used to model this decision stage.

5.2.2. The Second Decision Making Stage

The second stage of the decision making process selects and activates the most appropriate driving maneuver,

which is part of the set of feasible ones. Because the set of feasible driving maneuvers already contains only those maneuvers which can be safely performed in the specific road traffic situation, this stage is not safety-relevant, and its main objective is to maximize the efficiency and comfort.

Further details about the developed decision making approach have been published in [21, 22].

6. Experimental Decision Making Tests

The developed solution for the selection of feasible driving maneuvers has been implemented and integrated into the control software for an experimental autonomous vehicle. The identical control software is used for both, a real experimental vehicle and a simulated vehicle in a 3D simulation environment.

Experimental tests have been carried out using an autonomous vehicle (Cycab, manufactured by Robosoft, France), a second, manually driven Cycab and a conventional car. All vehicles, sensors, and test facilities were provided by the French research institute INRIA (team IMARA).

All three vehicles, including the conventional car, were equipped with differential GPS (DGPS) and were able to exchange data over a wireless network. Details about the sensors and the communication framework are omitted due to space limitations.

In addition to its own GPS position, the autonomous vehicle was able to receive the GPS positions of the other two vehicles. Furthermore, the autonomous vehicle’s World Model included a priori information, such as the position of intersections and positions of imaginary stop signs.

The main objective of the Cybercars-2 Communication Framework [23] is to enable autonomous vehicles to safely perform cooperative driving maneuvers.

The Cybercars-2 Communication Framework enables vehicles to send and receive data, regardless of whether they are autonomous or manually driven. Additionally,

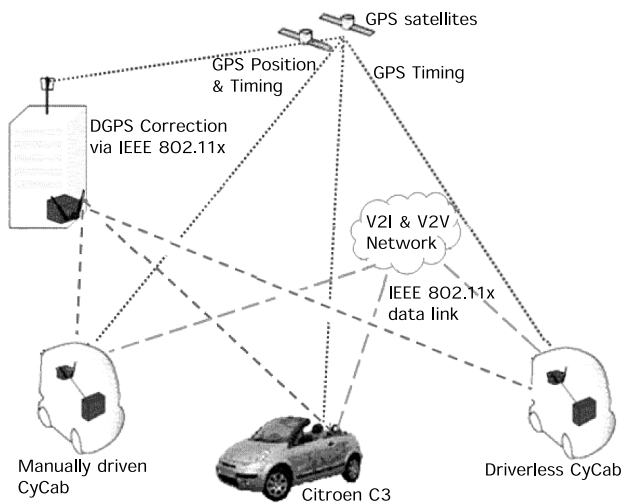


Fig. 5. Overview of the example communication setup (adapted from [23]). The autonomous CyCab communicated with a manually driven CyCab and a conventional car.

the communication framework concept includes communication with a traffic management centre. However, a detailed elaboration on the traffic management centre is beyond the scope of this work.

Figure 5 outlines the Cybercars-2 communication setup. The example setup consists of three communicating vehicles: an autonomous CyCab [24,25], a human-driven CyCab and a conventional car (Citroen C3) equipped with an Advanced Driver Assistance System. Advanced Driver Assistance Systems provide driving assistance functions, integrating telematic services and interaction between vehicles and the infrastructure [23]. CyCabs are computer controlled vehicles developed by INRIA and manufactured by Robosoft. They provide the option to be driven either autonomously or manually, using a joystick [24].

In order to test the decision making approach, three different traffic scenarios have been set up, all showing a common decision situation: passing a stopped vehicle under different traffic conditions.

In the first traffic scenario, the autonomous vehicle approached a stopped vehicle. The distance to the next intersection was sufficient to enable safe passing, and the oncoming traffic lane was free of any obstacles (**Fig. 6**). In this first scenario, the autonomous vehicle immediately started the passing maneuver when it approached the stopped vehicle.

The second traffic scenario was similar to the first, however another manually driven vehicle was oncoming, making safe passing unsafe. In this second scenario, the autonomous vehicle waited behind the stopped vehicle until the other vehicle passed, and started passing the stopped vehicle when the oncoming traffic lane was free.

In the third traffic scenario, a manually driven vehicle was stopped at an intersection. In this third traffic scenario, the autonomous vehicle waited behind the stopped



Fig. 6. First traffic scenario: the autonomous vehicle passes a stopped vehicle.

vehicle until it crossed the intersection. Then the autonomous vehicle continued driving, stopped at the imaginary stop sign and then continued crossing the intersection.

In the conducted experiments, the decision making module was repeatedly able to make the correct driving decision with respect to passing. Although it was not possible to continuously drive the vehicle for a longer distance due to a lack of adequate driving maneuvers, regarding only the correct decision making results, it can be concluded that the developed decision making approach is capable of dealing with real-world traffic conditions in real-time.

7. Conclusion

This paper has addressed the problem of real-time decision making for autonomous city vehicles, which is fundamental for enabling such vehicles to operate in non-simplified urban traffic conditions.

After giving an overview of the state of research, which however so far only includes solutions developed for simplified traffic conditions, the paper has presented the vehicle control system architecture, and addressed in more detail the three main subsystems relevant for decision making: the World Model, the Driving Maneuver subsystem, and the Real-Time Decision Making subsystem.

The main role of the World Model is to provide the Real-Time Decision Making subsystem with accurate and up-to-date information about the vehicle's traffic environment.

The Driving Maneuver subsystem represents the output results of decision making. Driving maneuvers are closed-loop control algorithms, each capable of maneuvering the autonomous vehicle in a specific traffic situation. Each driving maneuver is modeled based on a common finite automaton structure, which is capable of mod-

eling complex driving maneuvers, while enabling the decision making module to start and stop their execution. Furthermore, the developed finite automaton model enables the decomposition of complex driving maneuver control algorithms into subtasks with manageable complexity, and their implementation using closed-loop control methods in the Run states.

The purpose of the Real-Time Decision Making subsystem is to make driving decisions based on information from the World Model, and to activate the execution of the most appropriate driving maneuver. The task of decision making is decomposed into two consecutive stages. The first, safety-crucial decision making stage determines which driving maneuvers are feasible, i.e., safe to perform, conforming to traffic rules, and in line with the path planner indication.

The second decision making stage decides about the best execution alternative among the feasible driving maneuvers. The main focus of this stage is to improve comfort and efficiency with respect to competing, and possibly contradictory objectives.

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- Named the 2003 Queensland Professional Engineer of the Year by the Queensland Division of the Institution of Engineers Australia
- Awarded the 2004 Sir Lionel Hooke Award, by the Australian Council of the Institution of Engineering and Technology IET
- Awarded the 2004 IEE Achievement Medal (World-wide), IEE United Kingdom

Membership in Academic Societies:

- The Institution of Engineers Australia (EA)
 - The Institution of Engineering & Technology (IET)
 - The Institute of Electrical and Electronic Engineers (IEEE)
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